

Addressing Class Imbalance in MRI-based Alzheimer's Diagnosis using WGAN-GP

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Abstract—Alzheimer's Disease (AD) is a leading cause of dementia worldwide, and there is an urgent need for reliable early automated diagnostic tools. Among various automated diagnostic techniques, deep learning models, especially Convolutional Neural Networks (CNNs), have shown promising potential in classifying AD stages using Magnetic Resonance Imaging (MRI). However, the performance of these models depends on the dataset utilized for training. If the medical dataset is imbalanced, with samples from certain cases outnumbering others, this leads to biased model training and reduced sensitivity to minority classes. For instance, the popular OASIS-1 dataset has a ratio of 65:1 of non-demented to moderate dementia, respectively. Hence, most published work using this dataset fails to detect minority classes despite achieving high overall accuracy.

This study investigates the impact of generating realistic synthetic MRI slices using a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) for underrepresented classes. A lightweight CNN is trained on both the original imbalanced dataset and the balanced augmented dataset to assess the impact of WGAN-GP. The results obtained after training a CNN on the original OASIS-1 dataset and on a balanced dataset using WGAN-GP showed a recall of 100.00% for Moderate Dementia, up from 0.00% previously. Similarly, overall accuracy improved by 18.25% due to data balancing using WGAN-GP. These findings have huge implications in the field of medicine, where the performance of various machine learning (ML) and artificial intelligence (AI) classifiers can be improved through dataset balancing while simultaneously reducing the risk to patient privacy.

Index Terms—Alzheimer's Disease (AD), Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP), class imbalance, medical imaging, synthetic data

I. INTRODUCTION

Alzheimer's Disease (AD) is a chronic neurodegenerative condition that gradually impairs cognitive function and remains one of the most prevalent causes of dementia globally. According to a report published in the World Alzheimer Report, around 55 million people worldwide are diagnosed with AD, and this number is expected to reach 139 million by 2050 [1], [2]. Hence, there is an urgent need to develop a reliable,

scalable diagnostic solution. Additionally, early diagnosis of AD is critical for enabling timely clinical intervention before its progression to dementia. Besides various biomarkers used for diagnosing AD, including amyloid-beta ($A\beta$), tau proteins, and cerebrospinal fluid (CSF) biomarkers, Magnetic Resonance Imaging (MRI) is widely used as a non-invasive imaging modality to identify structural brain abnormalities associated with AD [3], [4]. In AD diagnosis using MRI, structural changes in brain anatomy are visually assessed, particularly hippocampal atrophy and ventricular enlargement. However, this requires manual examination of MRI scans, which is time-consuming and subject to clinicians' interpretation. Therefore, it becomes critical to develop automated diagnostic systems that are noninvasive and less dependent on subjective interpretation. In recent years, machine learning (ML) and artificial intelligence (AI), particularly deep learning techniques, have gained significant prominence in the automated diagnosis and classification of various diseases, including cancers and AD.

ML and AI have emerged as transformative tools for diagnosing and managing AD by leveraging data from diverse sources, including neuroimaging, genetic information, and electronic health records, to classify disease stages and support early detection. Saxena et al. used a dataset consisting of 150 individuals and their genetic and clinical records [5]. They investigated various ML classifiers, including Support Vector Machine (SVM), Decision Tree, Random Forest, Gaussian Naive Bayes, Gradient Boosting, and others. Among them, Random Forest and Gradient Boosting achieved an accuracy of 83.92%. In another study, ML classifiers such as Random Forest, XGBoost, and SVM were applied to the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, achieving an accuracy of 91% [6]. Similarly, deep learning techniques such as Convolutional Neural Networks (CNNs) were applied to the ADNI dataset to classify AD into various categories, achieving an impressive accuracy of 99.05% [7]. In a recent work, graph-based learning was explored in AD diagnosis,

where Graph Neural Network (GNN) models were used to analyze diffusion MRI (dMRI) and functional MRI (fMRI) data from the ADNI dataset [8]. Despite these promising advancements, the performance of many ML and deep learning models remains constrained by limitations such as limited dataset size and severe class imbalance.

Several publicly available datasets play a significant role in evaluating various ML and AI models for AD diagnosis. One of the most widely used datasets is ADNI [9], [10], and the Open Access Series of Imaging Studies (OASIS) [11], [12], including OASIS-1 and OASIS-2, are widely used due to their availability of structural MRI scans and associated clinical data. These datasets have made significant contributions to training and validating various automated diagnostic systems and advancing AI-based AD classification. Besides ADNI and OASIS datasets, other large-scale datasets, such as the UK Biobank and the National Alzheimer's Coordinating Center (NACC), have also been used to develop multimodal diagnostic models incorporating neuroimaging, genetic, and clinical features [13]. However, many studies using these datasets report high overall accuracy, which may not reflect true clinical performance. One of the primary reasons is an imbalance in class distribution across various AD datasets, where the samples corresponding to advanced dementia stages are significantly different from those of cognitively normal subjects. If left unaddressed, this leads to poor performance of automated diagnosis systems on minority classes, limiting their clinical applicability.

To address the limitations of imbalanced medical imaging datasets, generative models are a promising alternative for synthesizing realistic data samples. Among all generative models, Generative Adversarial Networks (GANs) have demonstrated superior potential for generating high-quality synthetic medical images that enhance model generalization. However, conventional GAN-based architectures often suffer from training instability and mode collapse, limiting their ability to produce diverse, reliable samples. Therefore, in this study, we investigate a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP)-based data augmentation framework to address class imbalance in MRI-based AD classification. The proposed framework generates synthetic MRI slices for underrepresented classes to improve minority-class representation during model training. The resulting balanced dataset is then evaluated using a lightweight CNN. Thus, this work is significant for developing a WGAN-GP-based data augmentation framework to address class imbalance in MRI-based AD classification and for evaluating its impact on class-wise and overall classification performance.

II. RELATED WORK

ML and deep learning techniques are widely used to classify AD using neuroimaging data. However, the performance of each technique depends heavily on the dataset's size, diversity, and class balance. Hence, in this section, we present various key developments in AD, with emphasis on deep learning

approaches, traditional methods for handling imbalance, and generative modeling techniques.

A. Deep Learning Approaches for Alzheimer's Disease Classification

Deep learning models, especially CNNs, have demonstrated strong performance in classifying AD across multiple classes using MRI data. In various studies, deep learning models such as ResNet, DenseNet, and VGG networks have been applied across multiple clinical stages using the ADNI [14] and OASIS [15] datasets.

A similar study was presented by Islam et al., in which a ResNet-based model was employed to classify MRI images from the OASIS dataset [16]. Though their study achieved an impressive accuracy of 93.18%, its performance was limited by the availability of annotated data, and only binary classification was performed, which restricts the model's ability to distinguish between intermediate dementia stages and limits its clinical applicability for multi-stage diagnosis. To improve multi-class classification performance, several studies applied deep CNN architectures, such as DenseNet, VGG, and 3D-CNN models, for multi-stage AD classification using MRI data. For instance, Feng et al. used a hybrid deep learning framework that integrated 3D-CNNs and recurrent layers for AD diagnosis, thereby improving performance across multiple stages [17]. Despite the use of advanced techniques that improve performance, multi-class classification remains a challenge due to the unequal distribution of MRI image samples across dementia stages. Due to these limitations, there is a critical need for effective strategies to address class imbalance in medical imaging datasets.

B. Traditional Techniques for Handling Class Imbalance

To mitigate the effects of imbalanced datasets, several algorithms and data-level techniques are proposed and employed, including cost-sensitive, weighted cross-entropy, and focal loss strategies [18]. In the cost-sensitive method, the loss function is modified so that each misclassification incurs a higher penalty for minority-class samples. Similarly, weighted cross-entropy and focal loss strategies have been widely adopted in medical image classification tasks to address imbalance-related bias. Another commonly utilized technique is oversampling using the Synthetic Minority Over-sampling Technique (SMOTE). For instance, Krishna et al. [19] applied SMOTE to a balanced MRI dataset, and after applying advanced deep learning techniques, ResNet-50, ResNet-152, and EfficientNet architectures, the precision and reliability of their multi-class classification performance improved significantly. However, SMOTE typically generates synthetic samples through interpolation in feature space rather than in image space. Hence, when applied to high-dimensional medical imaging data, it can lead to the generation of unrealistic anatomical structures and blurred textures, negatively affecting the classifier's performance. Furthermore, SMOTE does not preserve spatial coherence or anatomical realism, which is essential in neuroimaging-based diagnosis. These limitations highlight the

need for more advanced augmentation techniques capable of generating biologically meaningful data samples.

C. Generative Adversarial Networks for Medical Image Augmentation

In recent times, Generative Adversarial Networks (GANs) have emerged as an effective tool for generating synthetic images that resemble real-world anatomical structures. Unlike conventional oversampling techniques such as SMOTE, GANs learn from the underlying probability distribution of the training data and generate images that are much more diverse and realistic. In [20], Bowles et al. demonstrated the capabilities of GANs to generate realistic magnetic resonance (MR) images that simulate the progression of AD's anatomical changes. Hence, through their work, they highlighted the effectiveness of GANs in improving the ability of various ML and deep learning models to encode structural variations in advanced disease stages. However, GANs often suffer from instability, leading to mode collapse and the generation of similar images, thereby reducing diversity. These limitations reduce the effectiveness of GAN-based augmentation, particularly in applications that require high anatomical fidelity, such as brain MRI analysis.

D. Wasserstein GAN-Based Approaches

To overcome the limitations of traditional GAN-based architectures, Generative Adversarial Networks (WGANs) are introduced as a more stable alternative. Due to the use of Earth Mover's (Wasserstein) distance as a loss metric, which improves training stability and provides smoother gradient propagation during optimization, WGANs provide a more stable alternative for generating synthetic images than GANs. In [21], Murugan et al. utilized the WGANs to generate chest X-ray (CXR) images. Across various evaluation metrics, including Fréchet Inception Distance (FID) and Inception Score, they found that the synthetic images generated by WGANs were superior to those generated by the GAN, making them ideal for medical imaging research.

Hence, in this paper, we investigate the effectiveness of WGAN-GP architectures for generating high-quality synthetic MRI slices for minority classes to balance the AD-related OASIS-1 dataset. Additionally, after balancing the dataset, we examine its impact on class-wise classification and overall performance.

III. PROPOSED METHODOLOGY

The proposed methodology to address the severe class imbalance in the AD class through synthetic data generation includes data preprocessing, synthetic data generation using WGAN-GP, and validation using a lightweight CNN. This proposed pipeline ensures that the model learns across all dementia stages.

A. Dataset Description

To test the effectiveness of WGAN-GP, it is essential to use an imbalanced dataset; therefore, in our study, we use

MRI images from the OASIS-1 dataset, which is widely used for AD classification research. The dataset consists of T1-weighted structural MRI scans from different stages of dementia, categorized according to the Clinical Dementia Rating (CDR) scale. The dataset includes four primary diagnostic categories corresponding to CDR values of 0, 0.5, 1, and 2, representing cognitively normal subjects, very mild dementia, mild dementia, and moderate dementia, respectively. A major challenge in using this dataset for multi-class classification is its severe class imbalance, with the number of samples for cognitively normal subjects significantly exceeding that for moderate dementia cases. Hence, this can cause the model to prioritize the majority class, which in this dataset corresponds to cognitively normal subjects, undermining the goal of accurately identifying and classifying patients across all stages of dementia, particularly the clinically critical minority stages.

B. Data Preprocessing

Before training the generation and classification models, all MRI slices must undergo preprocessing to ensure consistency. In the first step, all the MRI images are resized to a fixed spatial resolution of 128×128 pixels. This step is essential to ensure uniform input dimensions across all samples and to reduce computational complexity without significantly affecting relevant anatomical structures. This step is followed by intensity normalization, in which pixel intensities are scaled to a standard range of $[0, 1]$ to improve numerical stability during training and facilitate faster convergence. Later, each MRI slice is assigned a class label based on its CDR value, and categorical labels are numerically encoded to enable supervised learning. Finally, due to the severe class imbalance in the original dataset, a custom stratified split was utilized to ensure a strict, unbiased evaluation. In this work, exactly 100 samples per class are isolated to create a perfectly balanced "Fair Test Set" (400 samples total). All remaining samples were used for the training set. Additionally, to avoid data leakage, subject-level separation was enforced prior to synthetic augmentation, and synthetic MRI slices were generated exclusively from the training set.

C. Synthetic Data Generation Using WGAN-GP

Synthetic MRI samples are generated using a WGAN-GP to address class imbalance. The WGAN-GP architecture consists of two competing neural networks. One is responsible for producing synthetic MRI images, while the other serves as a critic network that evaluates the authenticity of the generated images. Thus, the generator's role is to generate synthetic samples that resemble real MRI data. On the other hand, the critic learns to distinguish between real and generated samples, thereby improving the generator's performance over successive training iterations.

1) *Generator Architecture*: The generator network receives a latent noise vector sampled from a standard Gaussian distribution. The latent vector typically lies in a 100-dimensional space and serves as the initial representation from which synthetic MRI images are generated. The generator architecture

is divided into two parts. The initial part of the generator architecture consists of fully connected layers, whereas the latter part has multiple transposed convolutional layers that progressively increase spatial resolution. To stabilize training and introduce nonlinearity, batch normalization is applied between layers, and the Rectified Linear Unit (ReLU) activation function is used. In the end, the final output layer generates grayscale MRI images of size $1 \times 128 \times 128$ pixels. Hence, the primary objective of the generator is to learn the distribution of minority-class MRI samples and to generate realistic synthetic images that accurately represent underrepresented dementia stages.

2) *Critic Architecture*: The critic network takes both real and generated images as input and outputs a scalar value representing realism, unlike traditional GANs. The critic architecture consists of multiple convolutional layers, followed by Leaky ReLU activation functions and layer normalization to stabilize gradients during training. To prevent gradient saturation and improve training stability, a slope coefficient of 0.2 is used in the leaky ReLU activation function. A fully connected output layer produces the final scalar output used in computing the Wasserstein loss.

3) *WGAN-GP Loss Function*: The WGAN-GP framework utilizes the Wasserstein loss function combined with a gradient penalty to enforce Lipschitz continuity during training. The gradient penalty term improves stability by constraining the critic network's gradient norm. In this study, the gradient penalty coefficient is set to $\lambda = 10$, which ensures stable training and reduces the likelihood of mode collapse. This formulation enables the generator to produce diverse, realistic MRI samples that can help balance highly imbalanced datasets.

$$L = \mathbb{E}_{\tilde{x} \sim P_g} [D(\tilde{x})] - \mathbb{E}_{x \sim P_r} [D(x)] + \lambda \mathbb{E}_{\hat{x}} [(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2] \quad (1)$$

where P_r represents the real data distribution, P_g represents the generated data distribution, $P_{\hat{x}}$ denotes the distribution of interpolated samples, $D(x)$ denotes the critic output, and λ is the gradient penalty coefficient used to enforce Lipschitz continuity [22].

D. Classification Model for Validation

To evaluate the effectiveness of synthetic data generation, a lightweight vanilla CNN classifier is designed. In this work, we utilized a relatively simple CNN architecture, intentionally selected to ensure that improvements in classification performance are primarily due to dataset quality rather than increased model complexity.

The vanilla CNN architecture consists of three convolutional blocks designed for hierarchical feature extraction. The first convolutional layer contains 32 filters, followed by a second layer containing 64 filters and a third layer containing 128 filters. Each convolutional layer is followed by a ReLU activation function and a max-pooling layer to reduce spatial dimensions while retaining important spatial features. After feature extraction, fully connected layers are used to classify

input images into four dementia categories corresponding to the defined CDR classes.

To ensure fair and reliable performance comparison, evaluation metrics are computed using a balanced test dataset containing equal numbers of samples from each dementia class. The performance of the classification models is evaluated using multiple quantitative metrics, including accuracy, precision, recall (sensitivity), F1-score, and confusion matrix analysis. Among these metrics, special emphasis is placed on recall performance for minority classes, as this directly reflects WGAN-GP's ability to address class imbalance.

IV. RESULTS AND DISCUSSION

A. Baseline Performance on Imbalanced Dataset

The baseline vanilla CNN model is trained on the original OASIS-1 dataset, which is unbalanced and does not apply any augmentation techniques. As shown in Fig. 1, the baseline model exhibits lower convergence behavior compared to the WGAN-GP augmented model CNN trained on the balanced dataset. The loss curves reveal reduced training instability, while the validation accuracy trends confirm that dataset balancing leads to faster learning and consistently superior predictive performance. Additionally, the resulting confusion matrix shows a significant effect of class imbalance on the model's performance as shown in Fig 2. As shown in Table I, the baseline model achieves an overall test accuracy of 73.75%. However, a closer examination of the class-wise results in Table I reveals highly uneven performance across classes. While the Normal and Very Mild Dementia classes achieved strong F1-scores of 0.99 and 0.96, respectively, the Mild Dementia class achieved a low precision of 0.50 despite a recall of 0.96, resulting in an F1-score of 0.66. Most critically, the Moderate Dementia class achieved 0.00 precision, 0.00 recall, and 0.00 F1-score, indicating a complete failure to identify cases from this minority class. The results from the baseline dataset show that an imbalanced dataset can lead to biased learning towards the majority classes and poor performance for minority classes. Additionally, the model's inability to detect Moderate Dementia is particularly concerning in clinical applications, since accurate identification of advanced disease stages is essential for treatment planning and disease monitoring.

B. Performance After WGAN-GP-Based Dataset Balancing

To address the limitations observed in the baseline configuration, synthetic MRI slices were generated using the proposed WGAN-GP framework to balance the training set. Retraining the lightweight CNN on this balanced dataset yielded substantial performance gains. As shown in Table I, the overall test accuracy increased from the baseline of 73.75% to 92.00%, marking an improvement of 18.25%.

The primary objective, improving minority-class detection, was highly successful. The most significant advancement occurred in the Moderate Dementia (CDR 2) class, which progressed from a baseline recall of 0.00 to 1.00 following augmentation. This demonstrates a complete reversal of the

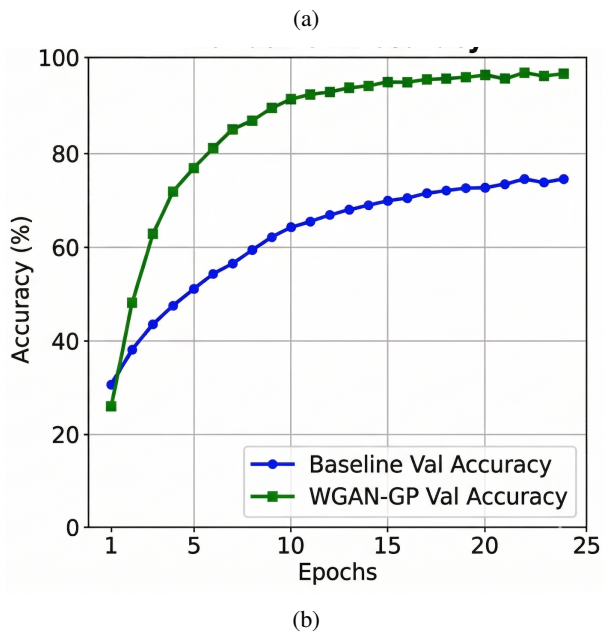
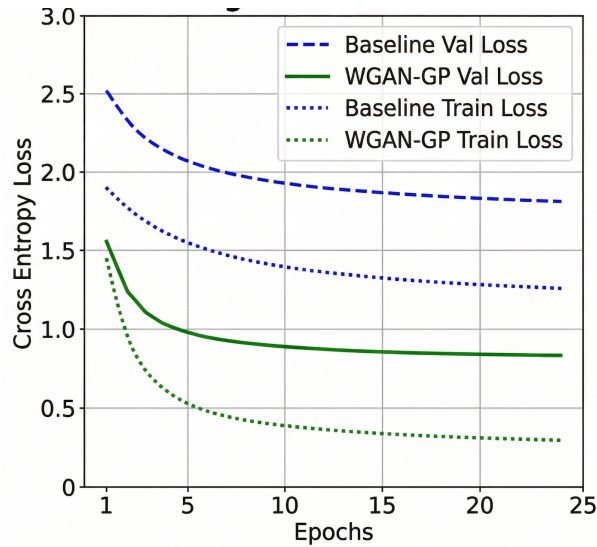
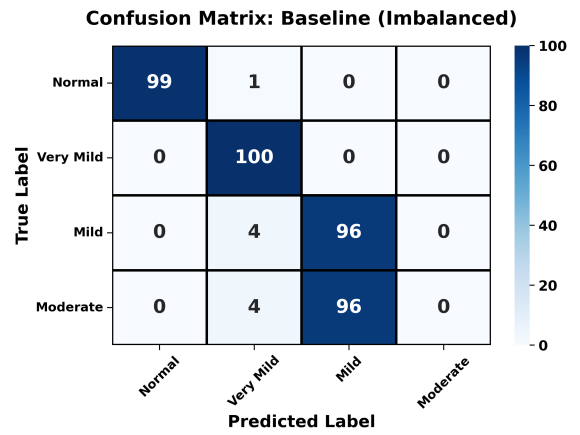
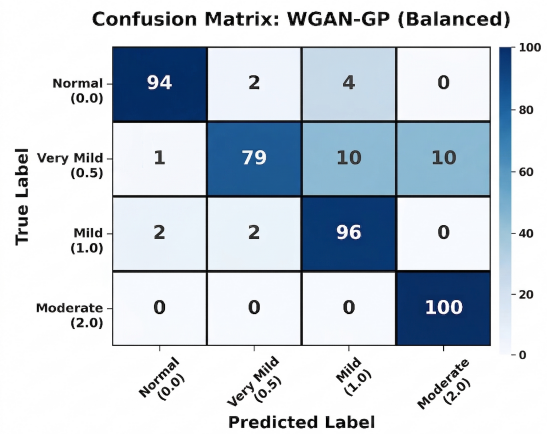


Fig. 1: Training performance comparison of the vanilla CNN model trained on the imbalanced dataset and the WGAN-GP balanced dataset. (a) Loss curves showing improved convergence behavior with dataset balancing. (b) Accuracy curves demonstrating significant performance improvement using WGAN-GP augmentation.

model's previous inability to identify advanced disease stages. Furthermore, performance stabilized for intermediate stages; Mild Dementia (CDR 1) achieved a robust recall of 0.96. Encouragingly, these gains in minority-class sensitivity did not substantially compromise majority-class performance. Normal cases (CDR 0) maintained high classification performance with a recall of 0.94. The macro-average metrics, with precision, recall, and F1-score all reaching 0.92, confirm that the WGAN-GP-based augmentation effectively fosters balanced learning across all dementia categories.



(a)



(b)

Fig. 2: Confusion matrices comparing classification performance of the vanilla CNN trained on (a) the original imbalanced dataset and (b) the WGAN-GP balanced dataset. Significant improvement in minority-class recognition is observed after dataset balancing.

C. Synthetic MRI Quality Assessment

Although the WGAN-GP model is designed to iteratively improve the quality of synthetic MRI images during training, it is also important to compare the generated images qualitatively with real MRI images. This comparison shows that the generated images accurately capture several key anatomical features observed in real MRI scans, including brain boundaries, ventricular regions, and cortical patterns, as shown in Fig. 3. These areas in the generated images are essential for meaningful classification tasks. Despite these promising observations, some synthetic MRI slices appeared slightly blurred or showed minor structural distortions. Additionally, the fine anatomical details were not consistently preserved across all the generated samples. Moreover, in some generated images, edge sharpness was reduced compared to the original MRI samples. Hence, these findings indicate that WGAN-GP-generated samples significantly improve classification perfor-

TABLE I: Performance comparison of the baseline CNN trained on the imbalanced dataset and the CNN trained on the WGAN-GP balanced dataset.

Model / Class	Precision	Recall	F1-score	Support
<i>Baseline CNN (Imbalanced Dataset)</i>				
Normal (0.0)	1.00	0.99	0.99	100
Very Mild (0.5)	0.92	1.00	0.96	100
Mild (1.0)	0.50	0.96	0.66	100
Moderate (2.0)	0.00	0.00	0.00	100
Accuracy		0.74		400
Macro Avg	0.61	0.74	0.65	400
Weighted Avg	0.61	0.74	0.65	400
<i>WGAN-GP CNN (Balanced Dataset)</i>				
Normal (0.0)	0.97	0.94	0.95	100
Very Mild (0.5)	0.95	0.79	0.86	100
Mild (1.0)	0.87	0.96	0.91	100
Moderate (2.0)	0.91	1.00	0.95	100
Accuracy		0.92		400
Macro Avg	0.93	0.92	0.92	400
Weighted Avg	0.93	0.92	0.92	400

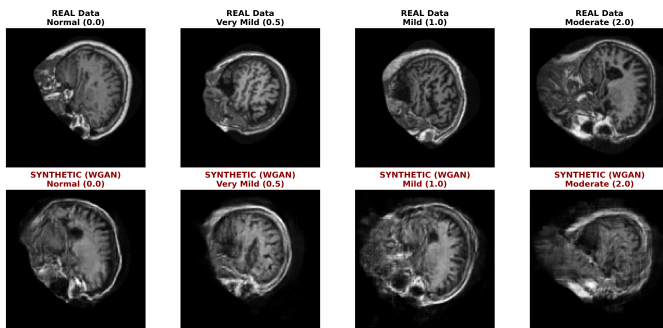


Fig. 3: Comparison of real and WGAN-GP-generated MRI slices across dementia stages.

mance, though the generated images are not always visually identical to real MRI scans.

D. Model Interpretability Using Grad-CAM

It is essential to understand how CNN models make classification decisions to assess the impact of the WGAN-GP-balanced dataset properly. For this purpose, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to visualize the spatial regions that influenced classification decisions, which is shown in Fig. 4. There is a clear difference in the Grad-CAM visualizations between the baseline-trained CNN model and the CNN model trained on the WGAN-GP-balanced dataset.

In the baseline model, activation maps were widely dispersed across the MRI images, with most of the attention allocated to peripheral regions that are not clinically relevant in a real-world scenario. In contrast, the activation maps for the WGAN-GP-augmented model show a more localized pattern in anatomically relevant regions associated with dementia progression. The activation maps also show increased consistency across classes, indicating improved feature learning. Overall, the Grad-CAM analysis shows that a balanced dataset not only

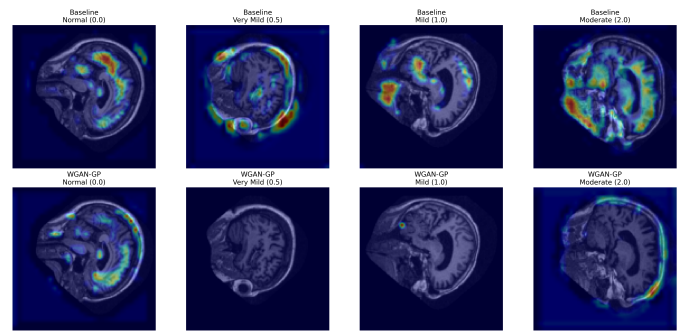


Fig. 4: Grad-CAM visualization comparing activation regions of the vanilla CNN trained on the imbalanced dataset and the WGAN-GP balanced dataset. The baseline model shows dispersed attention patterns, whereas the WGAN-GP-based model demonstrates more localized focus on clinically relevant brain regions associated with dementia progression.

improves classification performance but also enhances model interpretability by guiding the model toward clinically relevant anatomical features.

E. Limitations and Future Directions

Despite the use of WGAN-GP, which improved class-wise classification and overall performance, a few limitations were identified that warrant further investigation. Minor variations in the anatomical details of the generated samples may limit clinical interpretability and raise the need to investigate more advanced generative modeling techniques. In future work, alternative generative techniques such as diffusion models, variational autoencoders (VAEs), and conditional GAN architectures will be explored and compared with WGAN-GP to improve synthetic image fidelity.

V. CONCLUSION

This study addresses the critical challenge of severe class imbalance in medical imaging datasets by proposing a WGAN-GP-based data augmentation framework for MRI-based AD classification. By generating synthetic slices for underrepresented dementia stages in the OASIS-1 dataset, we significantly improved the diagnostic reliability of a lightweight CNN. Overall test accuracy increased by 18.25% (from 73.75% to 92.00%), and the model successfully identified clinically critical advanced disease stages, improving Moderate Dementia recall from 0.00 to 1.00. Furthermore, Grad-CAM analysis confirmed that the balanced dataset successfully guided the CNN to focus on clinically relevant anatomical regions rather than peripheral artifacts.

We acknowledge a limitation in the current generative approach: while the WGAN-GP model successfully captured the gross structural features required for classification, the synthetic MRI slices occasionally exhibited slightly lower visual fidelity, blurring, and edge degradation compared to real patient scans. However, despite these imperfections in raw image quality, training on this augmented data yielded vastly

superior and more clinically sound results than relying on the highly imbalanced baseline dataset.

Consequently, this work establishes a strong benchmark for generative augmentation in AD classification, demonstrating that mitigating minority-class underrepresentation can substantially improve diagnostic models even if the synthetic images are not yet perfectly photorealistic. Future research will build upon this benchmark by exploring more advanced generative architectures to further enhance synthetic image fidelity and clinical interpretability.

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REFERENCES

- [1] V. Viswan, N. Shaffi, M. Mahmud *et al.*, "Explainable artificial intelligence in alzheimer's disease classification: A systematic review," *Cognitive Computation*, vol. 16, no. 1, pp. 1–44, 2024.
- [2] Z.-P. Sang, T. Xue, Q.-R. Xing, Q.-Y. Zhang, H.-S. Chen, X. Wang, F.-R. Zhang, W.-L. Fu, W. Dong, S.-H. Huang, Y.-Y. Huang, and H.-B. Luo, "Discovery of -mangostin derivatives as novel pde4 inhibitors for the treatment of alzheimer's disease: An artificial intelligence-driven synergized strategy," *Chinese Chemical Letters*, vol. 37, no. 5, p. 111318, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1001841725005030>
- [3] N. M. Seidu, S. Kern, S. Sacuiu, T. R. Sterner, K. Blennow, H. Zetterberg, and I. Skoog, "Association of csf biomarkers with mri brain changes in alzheimer's disease," *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*, vol. 16, no. 1, p. e12556, 2024.
- [4] B. C. Dickerson, D. A. Wolk, and A. D. N. Initiative, "Biomarker-based prediction of progression in mci: comparison of ad signature and hippocampal volume with spinal fluid amyloid- β and tau," *Frontiers in Aging Neuroscience*, vol. 5, p. 55, 2013.
- [5] A. Saxena, Arpita, and S. Singh, "Detecting alzheimer's disease (ad) using machine learning algorithms," in *2024 4th Asian Conference on Innovation in Technology (ASIANCON)*, 2024, pp. 1–6.
- [6] M. Li, H. Liu, Y. Li, Z. Wang, Y. Yuan, and H. Dai, "Intelligent diagnosis of alzheimer's disease based on machine learning," in *Proceedings of the 2023 4th International Symposium on Artificial Intelligence for Medicine Science*, 2023, pp. 456–462.
- [7] P. K. Mandal and R. V. Mahto, "Deep multi-branch cnn architecture for early alzheimer's detection from brain mris," *Sensors*, vol. 23, no. 19, p. 8192, 2023.
- [8] K. Chen, Y. Weng, T. Dening, A. A. Hosseini, and G. Zuo, "Brain connectome graphs based machine learning for diagnosis of alzheimer's disease," *Alzheimer's & Dementia*, vol. 20, no. S1, p. e090237, 2024.
- [9] D. P. Veitch, M. W. Weiner, M. Miller, P. S. Aisen, M. A. Ashford, L. A. Beckett, R. C. Green, D. Harvey, C. R. Jack Jr, W. Jagust *et al.*, "The alzheimer's disease neuroimaging initiative in the era of alzheimer's disease treatment: a review ofadni studies from 2021 to 2022," *Alzheimer's & Dementia*, vol. 20, no. 1, pp. 652–694, 2024.
- [10] L. M. Shaw, M. Korecka, E. B. Lee, K. A. Cousins, H. Vanderstichele, S. E. Schindler, D. Tosun, M. L. DeMarco, M. Brylska, Y. Wan *et al.*, "Adni biomarker core: A review of progress since 2004 and future challenges," *Alzheimer's & Dementia*, vol. 21, no. 1, p. e14264, 2025.
- [11] D. S. Marcus, T. H. Wang, J. Parker, J. G. Csernansky, J. C. Morris, and R. L. Buckner, "Open access series of imaging studies (oasis): cross-sectional mri data in young, middle aged, nondemented, and demented older adults," *Journal of cognitive neuroscience*, vol. 19, no. 9, pp. 1498–1507, 2007.
- [12] M. A. Azam, K. B. Khan, M. Aqeel, A. R. Chishti, and M. N. Abbasi, "Analysis of the midas and oasis biomedical databases for the application of multimodal image processing," in *International Conference on Intelligent Technologies and Applications*. Springer, 2019, pp. 581–592.
- [13] Z. Yu, A. Mulholland, T. Huang, and Q. Liu, "Multimodal ai for alzheimer disease diagnosis: Systematic review of datasets, models, and modalities," *Journal of Medical Internet Research*, vol. 28, p. e85414, 2026.
- [14] A. Assmi, K. Elhabyb, A. Benba *et al.*, "Alzheimer's disease classification: a comprehensive study," *Multimedia Tools and Applications*, vol. 83, pp. 70 193–70 216, 2024.
- [15] P. K. Pandey, J. Pruthi, S. Alzahrani, A. Verma, and B. Zohra, "Enhancing healthcare recommendation: transfer learning in deep convolutional neural networks for alzheimer disease detection," *Frontiers in Medicine*, vol. 11, 2024.
- [16] J. Islam and Y. Zhang, "An ensemble of deep convolutional neural networks for alzheimer's disease detection and classification," *CoRR*, vol. abs/1712.01675, 2017. [Online]. Available: <http://arxiv.org/abs/1712.01675>
- [17] C. Feng, A. Elazab, P. Yang, T. Wang, F. Zhou, H. Hu, X. Xiao, and B. Lei, "Deep learning framework for alzheimer's disease diagnosis via 3d-cnn and fsbi-1stm," *IEEE Access*, vol. 7, pp. 63 605–63 618, 2019.
- [18] D. Zhou, S. Gao, and X. Huang, "Strategies for class-imbalanced learning in multi-sensor medical imaging," *Sensors*, vol. 26, no. 6, p. 1998, 2026.
- [19] K. S. Krishna, B. D. Kumar, M. D. Reddy, C. Saketh, and G. Ramasamy, "A multi-class classification framework with smote based data augmentation technique for alzheimer's disease progression," in *2024 International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE)*, 2024, pp. 1–6.
- [20] C. Bowles, R. Gunn, A. Hammers, and D. Rueckert, "Modelling the progression of alzheimer's disease in mri using generative adversarial networks," in *Medical Imaging 2018: Image Processing*, vol. 10574. SPIE, 2018, pp. 397–407.
- [21] K. Senthil Murugan and T. Swapna, "Synthetic data generation of chest x-ray images using modified wgan gp algorithm," in *International Conference on Data Science and Applications*. Springer, 2024, pp. 493–505.
- [22] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. C. Courville, "Improved training of wasserstein gans," *CoRR*, vol. abs/1704.00028, 2017. [Online]. Available: <http://arxiv.org/abs/1704.00028>